

QUESTION:13 CONSTRUCT A BINARY TREE

main.c

Share

Run

Output

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *left;
7      struct Node *right;
8  };
9
10 struct Node* createNode(int data) {
11     struct Node* node = (struct Node*)malloc(sizeof(struct Node));
12     node->data = data;
13     node->left = NULL;
14     node->right = NULL;
15     return node;
16 }
17
18 void printTree(struct Node *root, int space) {
19     if (root == NULL)
20         return;
21
22     space += 10;
23
24     printTree(root->right, space);
25
26     printf("\n");
27     for (int i = 10; i < space; i++)
28         printf(" ");
29     printf("%d\n", root->data);
30 }
```

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```
32 }
33
34 int main() {
35     struct Node *root = createNode(50);
36
37     root->left = createNode(30);
38     root->right = createNode(70);
39
40     root->left->left = createNode(20);
41     root->left->right = createNode(40);
42
43     root->right->left = createNode(60);
44     root->right->right = createNode(80);
45
46     root->left->left->left = createNode(10);
47     root->left->left->right = createNode(25);
48
49     root->left->right->left = createNode(35);
50     root->left->right->right = createNode(45);
51
52     root->right->left->left = createNode(55);
53     root->right->left->right = createNode(65);
54
55     root->right->right->left = createNode(75);
56     root->right->right->right = createNode(90);
57
58     printTree(root, 0);
59
60     return 0;
61 }
```

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=== Code Execution Successful ===